

GROAPA FINITA DE POTENTIAL. ALGORITMUL NEWTON-RAPHSON

ORIGIN := 1

$$h := 6.626026 \cdot 10^{-34}$$

$$m := 9.1 \cdot 10^{-31}$$

$$ee := 1.6 \cdot 10^{-19}$$

$$hp := \frac{h}{2 \cdot \pi}$$

Definirea gropii de potential

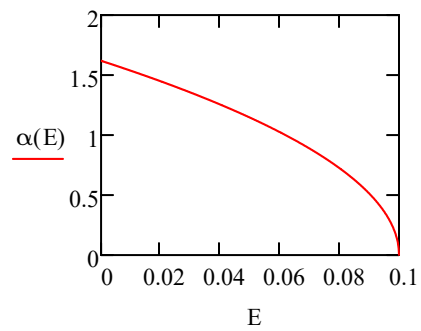
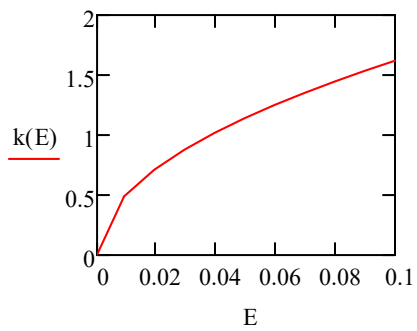
$$L := 10 \quad [\text{nm}]$$

$$V := 0.1 \quad [\text{eV}]$$

Calculul vectorului de unda

$$k(E) := \frac{\sqrt{2 \cdot m \cdot E \cdot ee}}{hp} \cdot 10^{-9}$$

$$\alpha(E) := \frac{\sqrt{2 \cdot m \cdot (V - E) \cdot ee}}{hp} \cdot 10^{-9}$$



$$f_p(E) := k(E) \cdot \tan\left(\frac{k(E) \cdot L}{2}\right) - \alpha(E) \quad \text{====> stari pare}$$

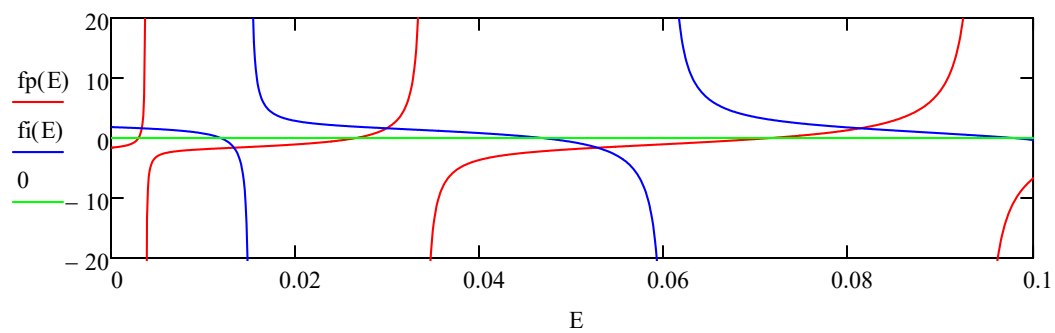
$$f_i(E) := k(E) \cdot \cot\left(\frac{k(E) \cdot L}{2}\right) + \alpha(E) \quad \text{====> stari impare}$$

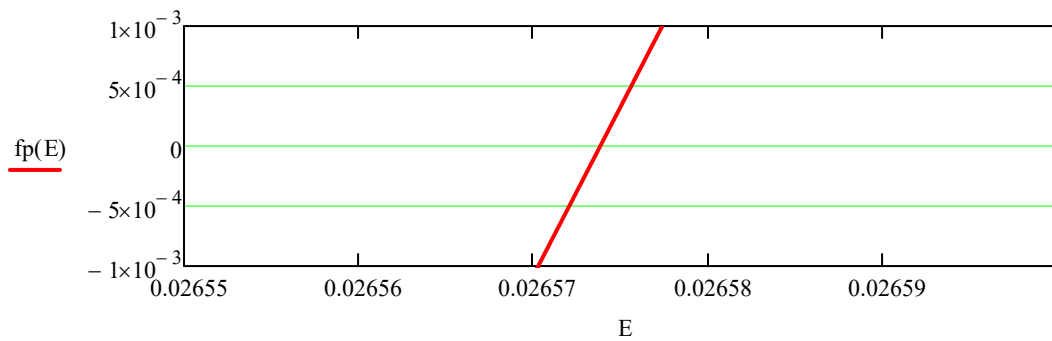
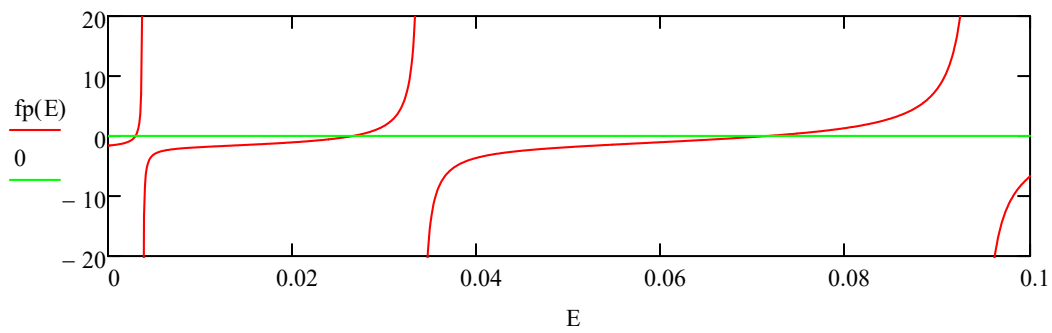
$$dk(E) := \frac{\sqrt{2 \cdot m \cdot ee}}{2 \cdot hp \sqrt{E}} \cdot 10^{-9}$$

$$d\alpha(E) := -\frac{\sqrt{2 \cdot m \cdot ee}}{2 \cdot hp \sqrt{(V - E)}} \cdot 10^{-9}$$

$$df_p(E) := \frac{d}{dE} f_p(E)$$

$$df_i(E) := \frac{d}{dE} f_i(E)$$





METODA NEWTON-RAPHSON

Determinarea energiei unei stari pana la o precizie de $\varepsilon = 10^{-6}$ eV

$$\varepsilon := 10^{-6}$$

Rulare cu "mana"

$$E_{p0} := 0.01$$

$$E_{p1} := E_{p0} - \frac{f_p(E_{p0})}{df_p(E_{p0})} \quad E_{p1} = 0.0318788275$$

$$E_{p2} := E_{p1} - \frac{f_p(E_{p1})}{df_p(E_{p1})} \quad E_{p2} = 0.0303670344$$

$$\text{Ep3} := \text{Ep2} - \frac{\text{fp}(\text{Ep2})}{\text{dfp}(\text{Ep2})} \quad \text{Ep3} = 0.0284773688$$

$$\text{Ep4} := \text{Ep3} - \frac{\text{fp}(\text{Ep3})}{\text{dfp}(\text{Ep3})} \quad \text{Ep4} = 0.0270362373$$

$$\text{Ep5} := \text{Ep4} - \frac{\text{fp}(\text{Ep4})}{\text{dfp}(\text{Ep4})} \quad \text{Ep5} = 0.0266001347$$

$$\text{Ep6} := \text{Ep5} - \frac{\text{fp}(\text{Ep5})}{\text{dfp}(\text{Ep5})} \quad \text{Ep6} = 0.0265739014$$

$$\text{Ep7} := \text{Ep6} - \frac{\text{fp}(\text{Ep6})}{\text{dfp}(\text{Ep6})} \quad \text{Ep7} = 0.0265738172$$

Aplicarie automatizata

Ep :=

```

aux1 ← 0.01
for i ∈ 2 .. 100
    auxi ← auxi-1 -  $\frac{\text{fp}(\text{aux}_{i-1})}{\text{dfp}(\text{aux}_{i-1})}$ 
    break if  $|\text{aux}_i - \text{aux}_{i-1}| < \varepsilon$ 
    ( continue ) otherwise
out1 ← i
out2 ← auxi
out
```

$$\text{Ep} = \begin{pmatrix} 8 \\ 0.0265738172 \end{pmatrix}$$

Ei :=

```

aux1 ← 0.02
for i ∈ 2 .. 1000
    auxi ← auxi-1 -  $\frac{\text{fi}(\text{aux}_{i-1})}{\text{dfi}(\text{aux}_{i-1})}$ 
    break if  $|\text{aux}_i - \text{aux}_{i-1}| < \varepsilon$ 
out1 ← i
out2 ← auxi
out
```

$$\text{Ei} = \begin{pmatrix} 7 \\ 0.0467253825 \end{pmatrix}$$